Title

Subtitle 1 Subtitle 2

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2026-03-04

Abstract

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1 Introduction

The devastation of rivers and streams from the Atlantic forest, biogeographical province (Udvardy, 1975) included in the list of the world's Hotspots (high biodiversity rates, endemism and anthropic pressure - Myers et al., 2000), is caused mainly by pollution, deforestation, damming and introduction of exotic species. Pollution decreases abundance and richness of loca0l fish, being aggravated by loss of riparian vegetation when there is deforestation or change of river course due to damming. These population reductions are also caused in the presence of invasive species, introduced accidentally or intentionally by humans, modifying the original competition/prey relation (Miranda, 2012). With about 2-5% of the original vegetation preserved in Atlantic forest, one of the main strategies for conserving and preserving biodiversity from this and other biogeographical provinces is the delimitation of areas known as Conservative Units (CU), regulated in Brazil by the National System of Conservative Unit (SNUC) and implemented by Chico Mendes Institute (ICMBio) to be managed at federal, state and local levels (Miranda, 2012; Arruda, 2014). Ecological data, such as richness and species abundance, are the most studied and represent how aquatic animals deal with general pollution, but the toxicity responsible for behavioral change depends on the organism, time of exposure and particle type, which demonstrates the need for physiological studies, specially in natural environments (Amoatey, Baawain, 2019). As physiological, morphological and behavioural aspects represent how their functions are performed in the ecosystem, they are nominated functional characteristics, commonly studied by parameters of strength and resilience. Changes on these parameters reflect environmental changes that could not be seen by parameters of taxonomical diversity, which happens when different species possess similar functions, highlighting how these, still scarce studies, can be promising (Júnior, 2021). Taking it into account, as also the relevance of Conservation Units for the Atlantic forest species, this study brings up the possibility to compare the ecological health from these waterbodies by identifying physiological stress markers from the fish within and near CU's. One of the main physiological functions affected by pollutants (such as pesticides, heavy metals, drugs and microplastics) is the osmoregulation one, that, as well as the pH control, is predominantly done by gills. Being also critical to gas exchange, ionic regulation and nitrogen excretion, its functions are partially performed by hormonal control of blood flow (carried out by catecholamines, which reduce vascular resistance in the gills and increase blood flow in the efferent artery via beta and alpha-adrenoreceptors, in that order) and mainly by the transport of ions, probably made in the chloride cells from the filament epithelium, that possess transport enzymes called sodium/potassium-dependent ATPases (Evans, 1987). The action of toxic compounds in the gills leads to a series of histopathologies in response to stress, such as necrosis and epithelial desquamation, frequently generated by flaws in cellular osmoregulation due to ATPase inhibition and physiological failure of the chloride cells or gills as a whole. The effects of this action on the organism can be measured through a few blood markers, mainly lower levels of Chlorine and Sodium ions, obtained by freshwater fish through exchange with internal ions (Evans, 1987), and higher levels of plasmatic proteins (Sabae, Mohamed, 2015). The hydration level of gills and muscles is another physiological parameter tied to the animal's capacity of dealing with salinity variation, and, for that, suffers mensurable effects from the action of toxic compounds in the environment. This active control parameter was suggested by Ayrapetyan (2012) as a universal biomarker for detecting pollution in the environment, being upheld by David et al. (2018) in a study with osmoconforming and euryhaline oysters under effect of different - but common - levels of salinity, and, more discretely, by Sabae and Mohamed (2015) in a study with freshwater fish. Another relevant biomarker is the abnormal activity of the multixenobiotic resistance (MXR) phenotype, an innate gene of many aquatic species that is expressed in organs such as gills, kidneys, blood-brain barrier, liver and pancreas. It is activated by exposure to pollutants and results in the production of P-glycoproteins (Pgp) in the membrane, which act to remove endogenous or exogenous toxic compounds out of the cell, protecting DNA and inducing its excretion. Although the MXR activity explains how some species are more resistant to the presence of pollutants, some substances can reverse its effectiveness, being called chemosensitizers (or simply MXR-inhibitors). Those compounds can cancel the Pgp activity, alter the mechanism regulation, saturate or even reverse MXR activity, acting, in this case, by saturating the Pgp pump and allowing other xenobiotics to enter the cell and increase its toxicity, even if the chemosensitizers themselves are not toxic. Precisely because they are mostly organic, natural, primarily harmless and very present in the materials emitted anthropogenic in nature, its detection should also be a priority in environmental risk studies (Smital, Kurelec, 1998). The hypothesis of markers linked to physiological stress of fish providing evidence of the environmental stress levels occurring inside and outside of a Conservation Unity was tested in the Guaribas Environmental Reserve (ReBio Guaribas), one of the 22 CU's inside the Atlantic forest area of Paraíba state, located in the Zona da Mata, wettest and most economically relevant mesoregion from the state (Fig. 1). Grouped, according to SNUC, as one of the integral protection areas (aimed to conserve biodiversity, being minority within the biogeographical province – 33,45% of the CU's), the definition of environmental reserves prohibits human interference inside their limits, except for management actions that aim its preservation and recovery. Despite being one of the most restricted types of Conservation Units, there are 22 populations that influence ReBio Guaribas due to historic factors of occupation and absence of an effective buffer zone around the three areas that make up the reserve. Those populations are mainly rural and cause direct impact, by using wood and plants from the reserve, and indirect, causing siltation of rivers and contaminating groundwater through the use of septic tanks and releasing agricultural chemicals into the soil, promoting possible comparable ecophysiological responses in the ichthyofauna inside and around the CU (Arruda, 2014; Ministério do Meio Ambiente, 2025). The ecophysiological evaluation of human impact on the forest, which is close to areas of monoculture and natural vegetation modified for livestock, was mainly guided by the fish species inventory of ReBio Guaribas made by Gouveia (2017), in a study that lists 18 species from five different orders: Characiformes (12 species), Perciformes (3 species), Synbranchiformes (1 species), Siluriformes (1 species) e Cyprinodontiformes (1 species). From those, two are exotic - Oreochromis niloticus, second most farmed fish species in the world and tolerant to osmotic variation, and Poecilia reticulata, largely used in mosquito control and resistant to stressful environments (Miranda, 2012) - and none are listed as threatened. The predominance of small fish inhabiting small basins explains the higher level of endemism in streams, resulting from the allopatric speciation generated by the low dispersion made by these fish and the consequent isolation of populations (Gouveia, 2017). Having in mind that these stream fish are phylogenetic close, this paper offers representative data for the entire local ichthyofauna using species of Hemigrammus genus, of the Characiformes order, characterized by a black conspicuous spot delimited on the dorsal fin. Its type species, Hemigrammus unilineatus, has a compressed and slightly elongated body and exhibits wide geographical distribution both in Central and South America, with registers in other Brazilian states, such as Alagoas and Pernambuco (Serra, 2010). The identification of physiological stress biomarkers in the collected fish aims to evaluate the ecological health of sites inside and surrounding the CU, through quantitative comparison of the influence of anthropogenic action with data from a legally protected area (statistical analysis made in the integrated programming environment R and R Studio - R Core Team, 2017; R Studio Team, 2022). This promotes a more tangible insight in contrast with other scientific papers that only make qualitative assessments of Conservative Units, and also using more specific markers than articles that only explore the ecological consequences of pollution exposure.

2 Material and Methods

Data and methods are discussed in Section 2.2. This citation will appear in the bibliography section (Gouveia et al. (2017); Medeiros et al. (2008)).

2.1 Study Area and Sampling Design

See publications by Gouveia et al. (2017).

2.2 Data Collection

See publications by Macêdo et al. (2019).

2.3 Statistical Analyses

Here you can add your statistical analyses and follow the text with the RStudio code. Following the model in https://elviomedeiros.github.io/Bentos2006_Q/.

```r
plot(1:10)
```

3 Results

The results section should follow the model in https://elviomedeiros.github.io/Bentos2006_Q/.

3.1 Environmental variables

3.2 Community structure

4 Discussion

Conclusions

Acknowledgements

Authorship contribution statement

Authorship of this paper is based on CRediT (2026).

Author1: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. Author2: Writing – review & editing, Data curation, Conceptualization. Author3: Writing – review & editing, Formal analysis, Data curation, Conceptualization.

Ethical approval

This study did not require ethical approval as it did not involve human participants or sensitive data. Field surveys were conducted based on collection permit number (IEF-10327-6 and SISBIO-10.635-7).

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Figures and Tables

Figures

Figure 1: Etapas do desenvolvimento da pesquisa científica.

Tables

Apendices

Non-used figures and tables

Non-used codes